中山大學 计算机网络实验报告

警示

1.实验报告如有雷同，雷同各方当次实验成绩均以 0 分计。
2.当次小组成员成绩只计学号、姓名登录在下表中的。
3.在规定时间内未上交实验报告的，不得以其他方式补交，当次成绩按 0 分计。
4.实验报告文件以 PDF 格式提交。

院系	计算机学院	班 级	计科 1 班	组长	陈安康
学号	22320021	22318001	22336070	22336064	
学生	陈安康	蔡可忻	冯乙山	杜雨彤	

编程实验

【实验内容】

(1) 完成路由器配置实验实例 7-3（P252）的"OSPF 单区域配置"，回答步骤 1、步骤 9 问题。

(2) 在（1）的基础上每台路由器上各加入一台电脑，画出新拓扑，然后：

(a) 检查任意两个 PC 之间是否可以 Ping 通，对一台主机 ping 其它主机的结果进行截屏。

(b) 采用#depug ip ospf 显示上面 OSPF 协议的运行情况，观察并保存 R1 发送和接收的 Update 分组(可以改变链路状态来触发)，注意其中 LSA 类型；观察有无 224.0.0.5、224.0.0.6 IP 地址，如有说明这两地址的作用。

(c) 显示并记录路由器 R1 数据库的 Router LSA，Network LSA，LS 数据库信息汇总

 # show ip ospf database router ！ 显示 router LSA

 # show ip ospf database network ！ 显示 network LSA

 # show ip ospf database database ！ 显示 OSPF 链路状态数据库信息。

(d) 显示并记录邻居状态。

 # show ip ospf neighbor

(e) 显示并记录 R1 的所有接口信息

#show ip ospf interface [接口名]

(f) 实验 7-4 (选做内容)

学号	学生	自评分
22320021	陈安康	100
22318001	蔡可忻	100
22336070	冯乙山	100
实验分工见个人心得		

【交实验报告】

上传实验报告：助教　　　　　　截止日期（不迟于）：1 周之内

上传包括两个文件：

（1）小组实验报告。上传文件名格式：小组号_ 编程实验.pdf （由组长负责上传）

例如: 文件名“10_ 编程实验.pdf”表示第 10 组的编程实验报告

（2）小组成员实验体会。每个同学单独交一份只填写了实验体会的实验报告。只需填写自己的学号和姓名。

文件名格式：小组号_学号_姓名_ 编程实验.pdf （由组员自行上传）

例如: 文件名“10_05373092_张三_ 编程实验.pdf”表示第 10 组的编程实验报告。

注意：不要打包上传！

步骤 1：

(1) 按照拓扑图配置 PC1 和 PC2 的 IP 地址、子网掩码、网关。使用 ping 命令，发现网络不连通，原因是还未进行相关的路由器配置。

```
C:\Users\D502>ping 192.168.5.11

正在 Ping 192.168.5.11 具有 32 字节的数据:
请求超时。
请求超时。
请求超时。
请求超时。

192.168.5.11 的 Ping 统计信息:
    数据包: 已发送 = 4，已接收 = 0，丢失 = 4 (100% 丢失)，

C:\Users\D502>
```

(2) 在路由器 R1 上执行 show ip route 命令，结果如下：

```
Password:
20-RSR20-1#show ip route

Codes:  C - connected, S - static, R - RIP, B - BGP
        O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default

Gateway of last resort is no set
20-RSR20-1#
```

因为还没有进行配置，所以路由表内容为空。

R2 路由表信息如下:

```
20-RSR20-2#show ip route

Codes:  C - connected, S - static, R - RIP, B - BGP
        O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default

Gateway of last resort is no set
20-RSR20-2#
```

步骤 9：

（1） 经过配置后，再次使用 show ip route 命令显示 R1 路由表信息如下:

```
20-RSR20-1#show ip route

Codes:  C - connected, S - static, R - RIP, B - BGP
        O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default

Gateway of last resort is no set
C    192.168.1.0/24 is directly connected, GigabitEthernet 0/1
C    192.168.1.1/32 is local host.
C    192.168.2.0/24 is directly connected, Serial 2/0
C    192.168.2.1/32 is local host.
O    192.168.3.0/24 [110/51] via 192.168.2.2, 00:00:20, Serial 2/0
O    192.168.5.0/24 [110/2] via 192.168.1.2, 00:00:58, GigabitEthernet 0/1
20-RSR20-1#
```

里面含有 O 条目，是通过互相传播 LSA 来学习产生的。比如其中一条 O 条目记录了目的网段的 IP，子网掩码为 192.168.3.0/24，下一跳的 IP 为 192.168.2.2，通过 Serial 2/0 端口转发。

R2 路由表信息如下:

```
20-RSR20-2#show ip route

Codes:  C - connected, S - static, R - RIP, B - BGP
        O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default

Gateway of last resort is no set
O    192.168.1.0/24 [110/51] via 192.168.2.1, 00:01:27, Serial 2/0
C    192.168.2.0/24 is directly connected, Serial 2/0
C    192.168.2.2/32 is local host.
C    192.168.3.0/24 is directly connected, GigabitEthernet 0/1
C    192.168.3.1/32 is local host.
O    192.168.5.0/24 [110/52] via 192.168.2.1, 00:01:27, Serial 2/0
20-RSR20-2#
```

交换机路由表信息如下：

```
20-S5750-1#
20-S5750-1#show ip route

Codes:  C - connected, S - static, R - RIP, B - BGP
        O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default

Gateway of last resort is no set
C     192.168.1.0/24 is directly connected, VLAN 10
C     192.168.1.2/32 is local host.
O     192.168.2.0/24 [110/51] via 192.168.1.1, 00:01:18, VLAN 10
O     192.168.3.0/24 [110/52] via 192.168.1.1, 00:00:31, VLAN 10
C     192.168.5.0/24 is directly connected, VLAN 50
C     192.168.5.1/32 is local host.
20-S5750-1#
```

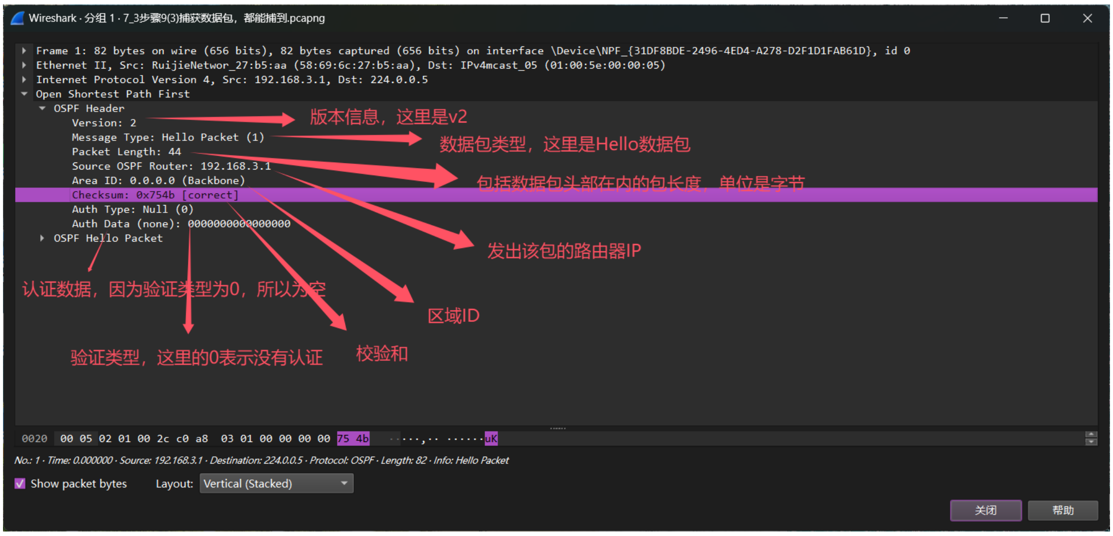
对比步骤 1 的路由表，会发现除了直连的网络的路由信息之外，还有通过 OSPF 学习到的路由信息，里面记录了目的网络的网段以及下一跳网络地址和转发接口。

（2） 执行 tracert 命令，结果如下：

```
C:\Users\D502>ping 192.168.5.11

正在 Ping 192.168.5.11 具有 32 字节的数据:
来自 192.168.5.11 的回复: 字节=32 时间=38ms TTL=125
来自 192.168.5.11 的回复: 字节=32 时间=36ms TTL=125
来自 192.168.5.11 的回复: 字节=32 时间=43ms TTL=125
来自 192.168.5.11 的回复: 字节=32 时间=35ms TTL=125

192.168.5.11 的 Ping 统计信息:
    数据包: 已发送 = 4, 已接收 = 4, 丢失 = 0 (0% 丢失),
往返行程的估计时间(以毫秒为单位):
    最短 = 35ms, 最长 = 43ms, 平均 = 38ms

C:\Users\D502>tracert 92.168.5.11
^C
C:\Users\D502>

C:\Users\D502>

C:\Users\D502>tracert 192.168.5.11

通过最多 30 个跃点跟踪
到 D52_12 [192.168.5.11] 的路由:

  1    <1 毫秒   <1 毫秒   <1 毫秒 192.168.3.1
  2    37 ms     37 ms     38 ms   192.168.2.1
  3    53 ms     50 ms     49 ms   192.168.1.2
  4    49 ms     49 ms     49 ms   D52_12 [192.168.5.11]

跟踪完成。
```

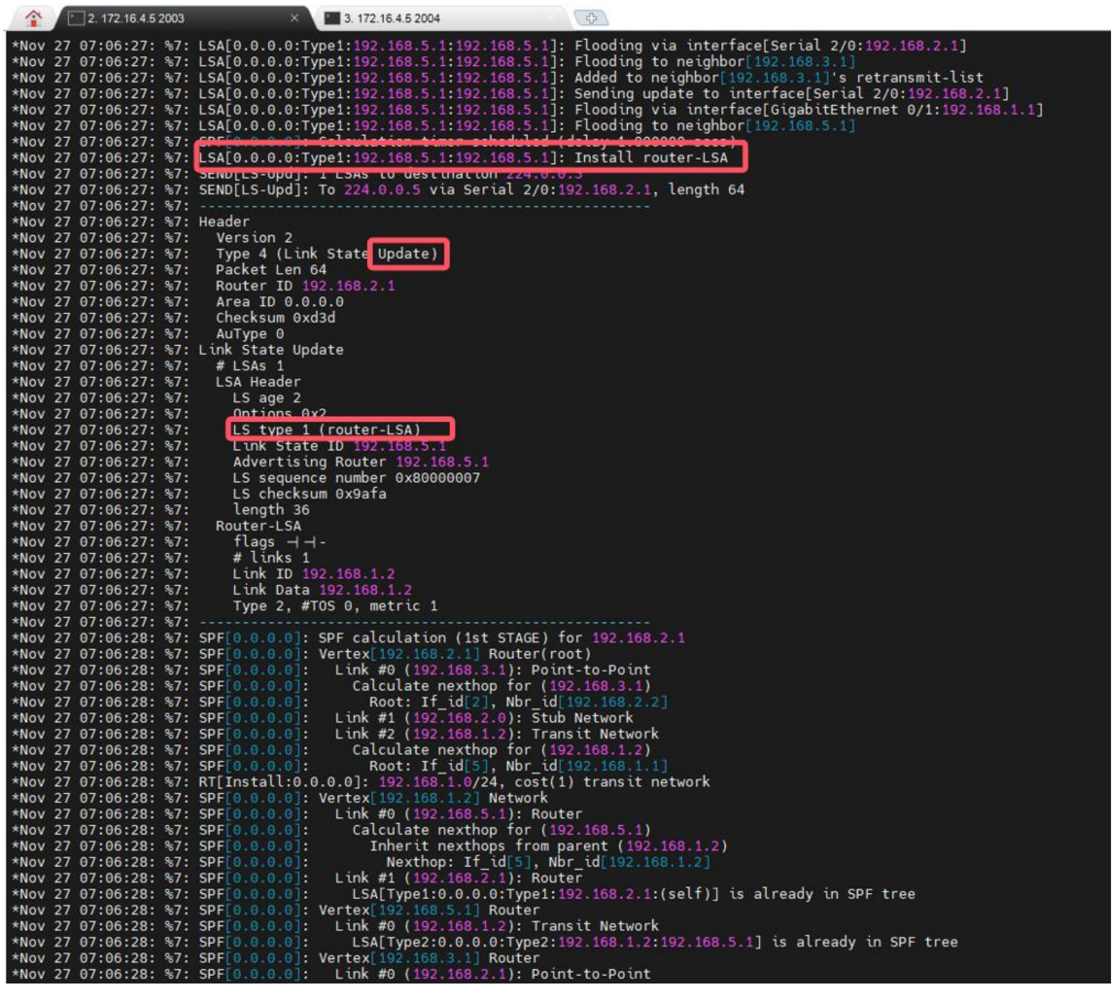
可以看到，数据经过路由器 R2 的 0/1 端口（192.168.3.1），通过 S2/0 端口到达路由器 R1 的 S2/0 端口（192.168.2.1）经过 R1 的 0/1 端口转发到交换机的 0/1 端口（192.168.1.2），最后到达目的主机（192.168.5.11）。链路由于 OSPF 的学习变得可以连通。

（3） 捕获数据包，分析 OSPF 数据包头部信息如下：



在 PC1 和 PC2 上都可以捕获到 OSPF 数据包。

（4） 使用 debug ip ospf 命令查看 debug 信息，在改变链路状态后获取到的 update 分组如下：




```
*Nov 27 07:06:27: %7: LSA[0.0.0.0:Type1:192.168.5.1:192.168.5.1]: Flooding via interface[Serial 2/0:192.168.2.1]
*Nov 27 07:06:27: %7: LSA[0.0.0.0:Type1:192.168.5.1:192.168.5.1]: Flooding to neighbor[192.168.3.1]
*Nov 27 07:06:27: %7: LSA[0.0.0.0:Type1:192.168.5.1:192.168.5.1]: Added to neighbor[192.168.3.1]'s retransmit-list
*Nov 27 07:06:27: %7: LSA[0.0.0.0:Type1:192.168.5.1:192.168.5.1]: Sending update to interface[Serial 2/0:192.168.2.1]
*Nov 27 07:06:27: %7: LSA[0.0.0.0:Type1:192.168.5.1:192.168.5.1]: Flooding via interface[GigabitEthernet 0/1:192.168.1.1]
*Nov 27 07:06:27: %7: LSA[0.0.0.0:Type1:192.168.5.1:192.168.5.1]: Flooding to neighbor[192.168.5.1]
*Nov 27 07:06:27: %7: SPF[0.0.0.0]: Calculation timer scheduled (delay 1.000000 secs)
*Nov 27 07:06:27: %7: LSA[0.0.0.0:Type1:192.168.5.1:192.168.5.1]: Install router-LSA
*Nov 27 07:06:27: %7: SEND[LS-Upd]: 1 LSAs to destination 224.0.0.5
*Nov 27 07:06:27: %7: SEND[LS-Upd]: To 224.0.0.5 via Serial 2/0:192.168.2.1, length 64
*Nov 27 07:06:27: %7: -----------------------------------------------------
*Nov 27 07:06:27: %7: Header
*Nov 27 07:06:27: %7:   Version 2
*Nov 27 07:06:27: %7:   Type 4 (Link State Update)
*Nov 27 07:06:27: %7:   Packet Len 64
*Nov 27 07:06:27: %7:   Router ID 192.168.2.1
*Nov 27 07:06:27: %7:   Area ID 0.0.0.0
*Nov 27 07:06:27: %7:   Checksum 0xd3d
*Nov 27 07:06:27: %7:   AuType 0
*Nov 27 07:06:27: %7: Link State Update
*Nov 27 07:06:27: %7:   # LSAs 1
*Nov 27 07:06:27: %7:   LSA Header
*Nov 27 07:06:27: %7:     LS age 2
*Nov 27 07:06:27: %7:     Options 0x2
*Nov 27 07:06:27: %7:     LS type 1 (router-LSA)
*Nov 27 07:06:27: %7:     Link State ID 192.168.5.1
*Nov 27 07:06:27: %7:     Advertising Router 192.168.5.1
*Nov 27 07:06:27: %7:     LS sequence number 0x80000007
*Nov 27 07:06:27: %7:     LS checksum 0x9afa
*Nov 27 07:06:27: %7:     length 36
*Nov 27 07:06:27: %7:   Router-LSA
*Nov 27 07:06:27: %7:     flags -|-|-
*Nov 27 07:06:27: %7:     # links 1
*Nov 27 07:06:27: %7:     Link ID 192.168.1.2
*Nov 27 07:06:27: %7:     Link Data 192.168.1.2
*Nov 27 07:06:27: %7:     Type 2, #TOS 0, metric 1
*Nov 27 07:06:27: %7: -----------------------------------------------------
*Nov 27 07:06:28: %7: SPF[0.0.0.0]: SPF calculation (1st STAGE) for 192.168.2.1
*Nov 27 07:06:28: %7: SPF[0.0.0.0]: Vertex[192.168.2.1] Router(root)
*Nov 27 07:06:28: %7: SPF[0.0.0.0]:   Link #0 (192.168.3.1): Point-to-Point
*Nov 27 07:06:28: %7: SPF[0.0.0.0]:     Calculate nexthop for (192.168.3.1)
*Nov 27 07:06:28: %7: SPF[0.0.0.0]:       Root: If_id[2], Nbr_id[192.168.2.2]
*Nov 27 07:06:28: %7: SPF[0.0.0.0]:   Link #1 (192.168.2.0): Stub Network
*Nov 27 07:06:28: %7: SPF[0.0.0.0]:   Link #2 (192.168.1.2): Transit Network
*Nov 27 07:06:28: %7: SPF[0.0.0.0]:     Calculate nexthop for (192.168.1.2)
*Nov 27 07:06:28: %7: SPF[0.0.0.0]:       Root: If_id[5], Nbr_id[192.168.1.1]
*Nov 27 07:06:28: %7: RT[Install:0.0.0.0]: 192.168.1.0/24, cost(1) transit network
*Nov 27 07:06:28: %7: SPF[0.0.0.0]: Vertex[192.168.1.2] Network
*Nov 27 07:06:28: %7: SPF[0.0.0.0]:   Link #0 (192.168.5.1): Router
*Nov 27 07:06:28: %7: SPF[0.0.0.0]:     Calculate nexthop for (192.168.5.1)
*Nov 27 07:06:28: %7: SPF[0.0.0.0]:       Inherit nexthops from parent (192.168.1.2)
*Nov 27 07:06:28: %7: SPF[0.0.0.0]:         Nexthop: If_id[5], Nbr_id[192.168.1.2]
*Nov 27 07:06:28: %7: SPF[0.0.0.0]:   Link #1 (192.168.2.1): Router
*Nov 27 07:06:28: %7: SPF[0.0.0.0]:     LSA[Type1:0.0.0.0:Type1:192.168.2.1:(self)] is already in SPF tree
*Nov 27 07:06:28: %7: SPF[0.0.0.0]: Vertex[192.168.5.1] Router
*Nov 27 07:06:28: %7: SPF[0.0.0.0]:   Link #0 (192.168.1.2): Transit Network
*Nov 27 07:06:28: %7: SPF[0.0.0.0]:     LSA[Type2:0.0.0.0:Type2:192.168.1.2:192.168.5.1] is already in SPF tree
*Nov 27 07:06:28: %7: SPF[0.0.0.0]: Vertex[192.168.3.1] Router
*Nov 27 07:06:28: %7: SPF[0.0.0.0]:   Link #0 (192.168.2.1): Point-to-Point
```

其中的 LS 类型为 1，为 Router LSA，表示该分组由每个路由器生成，描述了路由器连接的直连网络和链路信息，用于建立拓扑图和计算最短路径。只在本区域内传播。

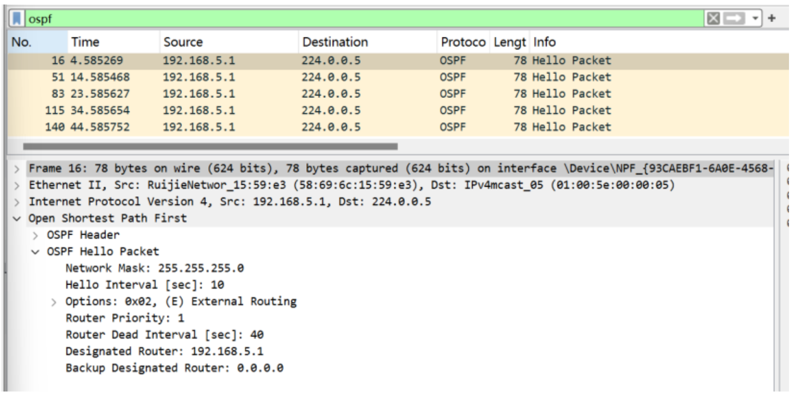
在上图中观察到了 224.0.0.5 IP 地址，没有观察到 224.0.0.6。这两个地址都是组播地址，其中 224.0.0.5 用于向所有 OSPF 路由器广播信息，而 224.0.0.6 则用于在多路访问网络中向 DR 和 BDR 发送特定的 OSPF 通信信息。

(5) 本实验中没有 DR/BDR。原因是本次实验的网络是点对点网络，每个节点都通过一条单独的物理链路直接连接到另一个节点，没有其他节点介入，不需要进行 DR 和 BDR 选举。

实验思考

（1）通过捕获的 ospf 抓包中可查看

Network Mask: 255.255.255.0



（3）子网掩码是 255.255.255.240，255.255.255.255-255.255.255.240=0.0.0.15

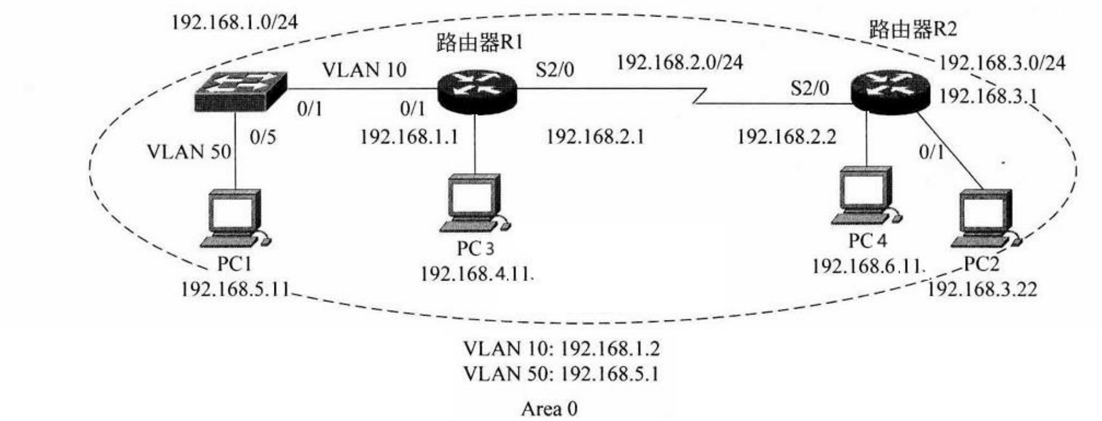
反掩码为 0.0.0.15

实验内容（2）

(1) 在（1）的基础上每台路由器上各加入一台电脑，画出新拓扑，然后：

(a) 检查任意两个 PC 之间是否可以 Ping 通，对一台主机 ping 其它主机的结果进行截屏。

绘制出新的拓扑图

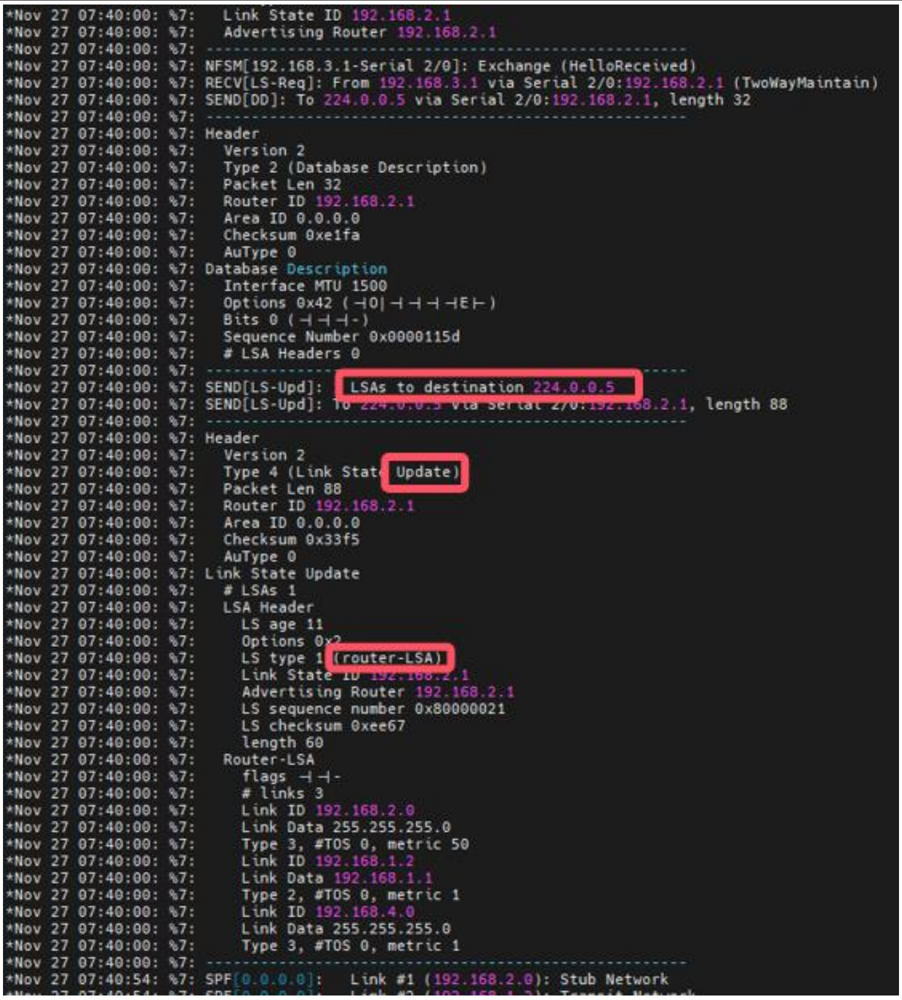


连接好电路后，进行连通性测试，192.168.4.11 ping 其他主机结果如下，可以连通

```
C:\Users\86135>ping 192.168.3.22

正在 Ping 192.168.3.22 具有 32 字节的数据:
来自 192.168.3.22 的回复: 字节=32 时间=39ms TTL=126
来自 192.168.3.22 的回复: 字节=32 时间=40ms TTL=126
来自 192.168.3.22 的回复: 字节=32 时间=39ms TTL=126
来自 192.168.3.22 的回复: 字节=32 时间=39ms TTL=126

192.168.3.22 的 Ping 统计信息:
    数据包: 已发送 = 4，已接收 = 4，丢失 = 0 (0% 丢失)，
往返行程的估计时间(以毫秒为单位):
    最短 = 39ms，最长 = 40ms，平均 = 39ms

C:\Users\86135>ping 192.168.5.11

正在 Ping 192.168.5.11 具有 32 字节的数据:
来自 192.168.5.11 的回复: 字节=32 时间=1ms TTL=126
来自 192.168.5.11 的回复: 字节=32 时间=1ms TTL=126
来自 192.168.5.11 的回复: 字节=32 时间=1ms TTL=126
来自 192.168.5.11 的回复: 字节=32 时间=1ms TTL=126

192.168.5.11 的 Ping 统计信息:
    数据包: 已发送 = 4，已接收 = 4，丢失 = 0 (0% 丢失)，
往返行程的估计时间(以毫秒为单位):
    最短 = 1ms，最长 = 1ms，平均 = 1ms

C:\Users\86135>ping 192.168.6.11

正在 Ping 192.168.6.11 具有 32 字节的数据:
来自 192.168.6.11 的回复: 字节=32 时间=36ms TTL=126
来自 192.168.6.11 的回复: 字节=32 时间=36ms TTL=126
来自 192.168.6.11 的回复: 字节=32 时间=39ms TTL=126
来自 192.168.6.11 的回复: 字节=32 时间=36ms TTL=126

192.168.6.11 的 Ping 统计信息:
    数据包: 已发送 = 4，已接收 = 4，丢失 = 0 (0% 丢失)，
往返行程的估计时间(以毫秒为单位):
    最短 = 36ms，最长 = 39ms，平均 = 36ms
```

(2)

（b）采用#depug ip ospf 显示上面 OSPF 协议的运行情况，观察并保存 R1 发送和接收的 Update 分组(可以改变链路状态来触发)，注意其中 LSA 类型；观察有无 224.0.0.5、224.0.0.6 IP 地址，如有说明这两地址的作用。



```
*Nov 27 07:40:00: %7:   Link State ID 192.168.2.1
*Nov 27 07:40:00: %7:   Advertising Router 192.168.2.1
*Nov 27 07:40:00: %7: --------------------------------------------------------
*Nov 27 07:40:00: %7: NFSM[192.168.3.1-Serial 2/0]: Exchange (HelloReceived)
*Nov 27 07:40:00: %7: RECV[LS-Req]: From 192.168.3.1 via Serial 2/0:192.168.2.1 (TwoWayMaintain)
*Nov 27 07:40:00: %7: SEND[DD]: To 224.0.0.5 via Serial 2/0:192.168.2.1, length 32
*Nov 27 07:40:00: %7: --------------------------------------------------------
*Nov 27 07:40:00: %7: Header
*Nov 27 07:40:00: %7:   Version 2
*Nov 27 07:40:00: %7:   Type 2 (Database Description)
*Nov 27 07:40:00: %7:   Packet Len 32
*Nov 27 07:40:00: %7:   Router ID 192.168.2.1
*Nov 27 07:40:00: %7:   Area ID 0.0.0.0
*Nov 27 07:40:00: %7:   Checksum 0xe1fa
*Nov 27 07:40:00: %7:   AuType 0
*Nov 27 07:40:00: %7: Database Description
*Nov 27 07:40:00: %7:   Interface MTU 1500
*Nov 27 07:40:00: %7:   Options 0x42 (-|O|-|-|-|-|E|-)
*Nov 27 07:40:00: %7:   Bits 0 (-|-|-|-)
*Nov 27 07:40:00: %7:   Sequence Number 0x0000115d
*Nov 27 07:40:00: %7:   # LSA Headers 0
*Nov 27 07:40:00: %7: --------------------------------------------------------
*Nov 27 07:40:00: %7: SEND[LS-Upd]:  LSAs to destination 224.0.0.5
*Nov 27 07:40:00: %7: SEND[LS-Upd]: To 224.0.0.5 via Serial 2/0:192.168.2.1, length 88
*Nov 27 07:40:00: %7: --------------------------------------------------------
*Nov 27 07:40:00: %7: Header
*Nov 27 07:40:00: %7:   Version 2
*Nov 27 07:40:00: %7:   Type 4 (Link State Update)
*Nov 27 07:40:00: %7:   Packet Len 88
*Nov 27 07:40:00: %7:   Router ID 192.168.2.1
*Nov 27 07:40:00: %7:   Area ID 0.0.0.0
*Nov 27 07:40:00: %7:   Checksum 0x33f5
*Nov 27 07:40:00: %7:   AuType 0
*Nov 27 07:40:00: %7: Link State Update
*Nov 27 07:40:00: %7:   # LSAs 1
*Nov 27 07:40:00: %7:   LSA Header
*Nov 27 07:40:00: %7:     LS age 11
*Nov 27 07:40:00: %7:     Options 0x2
*Nov 27 07:40:00: %7:     LS type 1 (router-LSA)
*Nov 27 07:40:00: %7:     Link State ID 192.168.2.1
*Nov 27 07:40:00: %7:     Advertising Router 192.168.2.1
*Nov 27 07:40:00: %7:     LS sequence number 0x80000021
*Nov 27 07:40:00: %7:     LS checksum 0xee67
*Nov 27 07:40:00: %7:     length 60
*Nov 27 07:40:00: %7:   Router-LSA
*Nov 27 07:40:00: %7:     flags -|-|-
*Nov 27 07:40:00: %7:     # links 3
*Nov 27 07:40:00: %7:     Link ID 192.168.2.0
*Nov 27 07:40:00: %7:     Link Data 255.255.255.0
*Nov 27 07:40:00: %7:     Type 3, #TOS 0, metric 50
*Nov 27 07:40:00: %7:     Link ID 192.168.1.2
*Nov 27 07:40:00: %7:     Link Data 192.168.1.1
*Nov 27 07:40:00: %7:     Type 2, #TOS 0, metric 1
*Nov 27 07:40:00: %7:     Link ID 192.168.4.0
*Nov 27 07:40:00: %7:     Link Data 255.255.255.0
*Nov 27 07:40:00: %7:     Type 3, #TOS 0, metric 1
*Nov 27 07:40:00: %7: --------------------------------------------------------
*Nov 27 07:40:54: %7: SPF[0.0.0.0]:   Link #1 (192.168.2.0): Stub Network
```

实验过程中有观察到发出和接收的路由信息更新通告数据包。更新路由的周期大约是 10 秒。在路由信息中有 LSA 类型，其中记录了路由器的链路状态信息，有观察到 224.0.0.5 的 IP 地址，没有观察到 224.0.0.6 的 IP 地址，这两个地址都是组播地址，其中 224.0.0.6 是 DR/BDR 发送给其他路由器的组播地址，224.0.0.5 是非 DR/BDR 路由器发送给其他路由器的组播地址。

（c）显示并记录路由器 R1 数据库的 Router LSA，Network LSA，LS 数据库信息汇总

 # show ip ospf database router ！ 显示 router LSA

```
20-RSR20-1#show ip ospf database router

            OSPF Router with ID (192.168.2.1) (Process ID 1)

                Router Link States (Area 0.0.0.0)

  LS age: 298
  Options: 0x2 (-|-|-|-|-|-|E|-)
  Flags: 0x0
  LS Type: router-LSA
  Link State ID: 192.168.2.1
  Advertising Router: 192.168.2.1
  LS Seq Number: 80000026
  Checksum: 0xbc7a
  Length: 72
   Number of Links: 4

    Link connected to: another Router (point-to-point)
     (Link ID) Neighboring Router ID: 192.168.3.1
     (Link Data) Router Interface address: 192.168.2.1
      Number of TOS metrics: 0
       TOS 0 Metric: 50

    Link connected to: Stub Network
     (Link ID) Network/subnet number: 192.168.2.0
     (Link Data) Network Mask: 255.255.255.0
      Number of TOS metrics: 0
       TOS 0 Metric: 50

    Link connected to: a Transit Network
     (Link ID) Designated Router address: 192.168.1.2
     (Link Data) Router Interface address: 192.168.1.1
      Number of TOS metrics: 0
       TOS 0 Metric: 1

    Link connected to: Stub Network
     (Link ID) Network/subnet number: 192.168.4.0
     (Link Data) Network Mask: 255.255.255.0
      Number of TOS metrics: 0
       TOS 0 Metric: 1

  LS age: 299
  Options: 0x2 (-|-|-|-|-|-|E|-)
  Flags: 0x0
  LS Type: router-LSA
  Link State ID: 192.168.3.1
  Advertising Router: 192.168.3.1
  LS Seq Number: 8000001c
  Checksum: 0xe5c1
  Length: 72
   Number of Links: 4

    Link connected to: another Router (point-to-point)
     (Link ID) Neighboring Router ID: 192.168.2.1
     (Link Data) Router Interface address: 192.168.2.2
      Number of TOS metrics: 0
       TOS 0 Metric: 50

    Link connected to: Stub Network
     (Link ID) Network/subnet number: 192.168.2.0
     (Link Data) Network Mask: 255.255.255.0
      Number of TOS metrics: 0
       TOS 0 Metric: 50

    Link connected to: Stub Network
     (Link ID) Network/subnet number: 192.168.3.0
     (Link Data) Network Mask: 255.255.255.0
      Number of TOS metrics: 0
       TOS 0 Metric: 1

    Link connected to: Stub Network
     (Link ID) Network/subnet number: 192.168.6.0
     (Link Data) Network Mask: 255.255.255.0
      Number of TOS metrics: 0
       TOS 0 Metric: 1

  LS age: 698
  Options: 0x2 (-|-|-|-|-|-|E|-)
  Flags: 0x0
  LS Type: router-LSA
  Link State ID: 192.168.5.1
  Advertising Router: 192.168.5.1
  LS Seq Number: 8000000c
  Checksum: 0xf31d
  Length: 48
   Number of Links: 2

    Link connected to: Stub Network
     (Link ID) Network/subnet number: 192.168.5.0
```

 # show ip ospf database network ！显示 network LSA

```
20-RSR20-1#show ip ospf database network

            OSPF Router with ID (192.168.2.1) (Process ID 1)

                Network Link States (Area 0.0.0.0)

  LS age: 837
  Options: 0x2 (-|-|-|-|-|-|E|-)
  LS Type: network-LSA
  Link State ID: 192.168.1.2 (address of Designated Router)
  Advertising Router: 192.168.5.1
  LS Seq Number: 80000003
  Checksum: 0x910f
  Length: 32
  Network Mask: /24
        Attached Router: 192.168.5.1
        Attached Router: 192.168.2.1
```

 # show ip ospf database database ！显示 OSPF 链路状态数据库信息。

```
20-RSR20-1#show ip ospf database database

OSPF process 1:

Area 0.0.0.0 database summary:
Router Link States         : 3
Network Link States        : 1
Summary Link States        : 0
ASBR-Summary Link States   : 0
NSSA-external Link States: 0
Link-Local Opaque-LSA      : 0
Area-Local Opaque-LSA      : 0
Total LSA                  : 4

Process 1 database summary:
Router Link States         : 3
Network Link States        : 1
Summary Link States        : 0
ASBR-Summary Link States   : 0
AS External Link States    : 0
NSSA-external Link States: 0
Link-Local Opaque-LSA      : 0
Area-Local Opaque-LSA      : 0
AS-Global Opaque-LSA       : 0
Total LSA                  : 4
```

（d）显示并记录邻居状态。

 # show ip ospf neighbor

```
20-RSR20-1#show ip ospf neighbor

OSPF process 1, 2 Neighbors, 2 is Full:
Neighbor ID     Pri   State                BFD State  Dead Time   Address         Interface
192.168.5.1       1   Full/DR              -          00:00:38    192.168.1.2     GigabitEthernet
0/1
192.168.3.1       1   Full/ -              -          00:00:40    192.168.2.2     Serial 2/0
```

（e）显示并记录 R1 的所有接口信息

#show ip ospf interface [接口名]

```
20-RSR20-1#show ip ospf interface gigabitEthernet 0/1
GigabitEthernet 0/1 is up, line protocol is up
  Internet Address 192.168.1.1/24, Ifindex 5, Area 0.0.0.0, MTU 1500
  Matching network config: 192.168.1.0/24
  Process ID 1, Router ID 192.168.2.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 192.168.5.1, Interface Address 192.168.1.2
  Backup Designated Router (ID) 192.168.2.1, Interface Address 192.168.1.1
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:06
  Neighbor Count is 1, Adjacent neighbor count is 1
  Crypt Sequence Number is 0
  Hello received 387 sent 389, DD received 11 sent 12
  LS-Req received 3 sent 1, LS-Upd received 11 sent 48
  LS-Ack received 40 sent 10, Discarded 0

20-RSR20-1#show ip ospf interface gigabitEthernet 0/0
GigabitEthernet 0/0 is up, line protocol is up
  Internet Address 192.168.4.1/24, Ifindex 4, Area 0.0.0.0, MTU 1500
  Matching network config: 192.168.4.0/24
  Process ID 1, Router ID 192.168.2.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 192.168.2.1, Interface Address 192.168.4.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:09
  Neighbor Count is 0, Adjacent neighbor count is 0
  Crypt Sequence Number is 0
  Hello received 0 sent 133, DD received 0 sent 0
  LS-Req received 0 sent 0, LS-Upd received 0 sent 0
  LS-Ack received 0 sent 0, Discarded 0

20-RSR20-1#show ip ospf interface se
20-RSR20-1#show ip ospf interface serial 2/0
Serial 2/0 is up, line protocol is up
  Internet Address 192.168.2.1/24, Ifindex 2, Area 0.0.0.0, MTU 1500
  Matching network config: 192.168.2.0/24
  Process ID 1, Router ID 192.168.2.1, Network Type POINTOPOINT, Cost: 50
  Transmit Delay is 1 sec, State Point-To-Point
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:03
  Neighbor Count is 1, Adjacent neighbor count is 1
  Crypt Sequence Number is 0
  Hello received 376 sent 419, DD received 31 sent 39
  LS-Req received 8 sent 14, LS-Upd received 27 sent 44
  LS-Ack received 42 sent 27, Discarded 0
```